SHEAFIFICATION PRESERVES ADEQUACY OF A PRESHEAF OF MODULES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The plus construction and the description of sheafification. Sites, Section sites-section-sheafification, invoked at proof lines 1996-2003. Fixes the object to be computed and its double-plus description.
- The flat base change property of adequate functors. Same chapter, Lemma lemma-adequate-flat, invoked at proof lines 2008-2014. Turns the Cech H^0 into a kernel of products of F(A) tensor A_i.
- Exactness of faithfully flat descent for modules. Descent, Lemma descent-lemma-ff-exact, invoked at proof lines 2015-2017. Identifies that kernel with F(A).

Source reference. Original lemma 023M: sources/stacks-chow/tags.txt, source lines 2747–2747. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06UW: sources/stacks-adequate/pinned-original-source.tex, source lines 442–442. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 00W1: sources/stacks-chow/tags.txt, source lines 1167–1167. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{F}$ be a presheaf of $\mathcal{O}$-modules on $(Sch/S)_\tau$. If for every affine scheme $\mathrm{Spec}(A)$ over S the functor $F_{\mathcal{F},A}$ is adequate, then the sheafification of $\mathcal{F}$ is an adequate $\mathcal{O}$-module.*

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Proof. Let $U = \mathrm{Spec}(A)$ be an affine scheme over S. Set $F = F_{\mathcal{F},A}$. The sheafification $\mathcal{F}^\# = (\mathcal{F}^+)^+$, see Sites, Section 00W1. By construction

$$(\mathcal{F})^+(U) = \mathrm{colim}_{\mathcal{U}} \check{H}^0(\mathcal{U}, \mathcal{F})$$

where the colimit is over coverings in the site $(Sch/S)_\tau$. Since U is affine it suffices to take the limit over standard affine τ-coverings $\mathcal{U} = \{U_i \to U\}_{i\in I} = \{\mathrm{Spec}(A_i) \to \mathrm{Spec}(A)\}_{i\in I}$ of U. Since each $A \to A_i$ and $A \to A_i \otimes_A A_j$ is flat we see that

$$\check{H}^0(\mathcal{U}, \mathcal{F}) = \mathrm{Ker}(\prod F(A) \otimes_A A_i \to \prod F(A) \otimes_A A_i \otimes_A A_j)$$

by Lemma 06UW. Since $A \to \prod A_i$ is faithfully flat we see that this always is canonically isomorphic to $F(A)$ by Descent, Lemma 023M. Thus the presheaf $(\mathcal{F})^+$ has the same value as $\mathcal{F}$ on all affine schemes over S. Repeating the argument once more we deduce the same thing for $\mathcal{F}^\# = ((\mathcal{F})^+)^+$. Thus $F_{\mathcal{F},A} = F_{\mathcal{F}^\#,A}$ and we conclude that $\mathcal{F}^\#$ is adequate. □